

Question 1 :

Sr No.	Error	Line Number	Error Type	Error Correction
1	Missing semicolon after array initialization	6	Compile-Time Error	Write int marks[5] = {80,90,70,60,50};
2	Loop condition i <= 5 accesses marks[5]	12	Runtime Error	Replace with for(int i = 0; i < 5; i++).
3	Integer division used for average calculation	20	Logical Error	Use average = total / 5.0;.
4	%d used to print a float variable	22	Compile-Time Warning	Use printf("Average = %.2f\n", average);.
5	scores[5] is out of bounds	33	Runtime Error	Valid indices are 0 to 4.
6	Buffer overflow in strcpy(subject,"Programming")	45	Runtime Error	Increase array size or use a shorter string.
7	String "one" used as an array index	53	Compile-Time Error	Use an integer index, e.g., attendance[1] = 1;.
8	Floating-point value 2.5 used as an array index	55	Compile-Time Error	Use an integer index.
9	Loop condition i <= 5 accesses attendance[5]	63	Runtime Error	Replace with for(int i = 0; i < 5; i++).
10	scanf() missing address operator &	74	Runtime Error	Use scanf("%d", &arr[i]);.
11	String assigned to character element	88	Compile-Time Error	Replace with grade[0] = 'A';.
12	Character array is not null-terminated before printing as a string	88–91	Runtime Error	Add grade[2] = '\0';.

Sr No.	Error	Line Number	Error Type	Error Correction
13	Assignment operator = used instead of comparison operator ==	97	Logical Error	Replace with <code>if(prices[2] == 300).</code>
14	Entire array assigned to another array	107	Compile-Time Error	Copy elements individually or use <code>memcpy()</code> .
15	%d used to print a float value	115	Compile-Time Warning	Use <code>printf("%.2f\n", salary[0]);</code> .
16	String assigned directly to character array	123	Compile-Time Error	Use <code>strcpy(city, "Jaipur");</code> .
17	Loop condition <code>i <= 5</code> accesses <code>result[5]</code>	131	Runtime Error	Replace with <code>for(int i = 0; i < 5; i++)</code> .
18	Possible undefined value in grade due to incomplete initialization	88–91	Runtime Error	Properly initialize the string, e.g., <code>char grade[] = "A+";</code> .
19	Array index out of bounds occurs in multiple arrays (marks, scores, attendance, result)	12, 33, 63, 131	Runtime Error	Ensure all indices remain between 0 and 4.
20	Program contains multiple cases of undefined behavior due to invalid array access	Multiple Lines	Runtime Error	Validate indices and use correct array bounds checks.

Question 2 :

Sr No.	Error	Line Number	Error Type	Error Correction
1	Loop condition <code>i <= 10</code> accesses <code>numbers[10]</code>	16	Runtime Error	Replace with <code>for(int i = 0; i < 10; i++)</code> .
2	Sum is overwritten instead of accumulated	27	Logical Error	Replace with <code>sum += numbers[i];</code> .

Sr No.	Error	Line Number	Error Type	Error Correction
3	Integer division used for average calculation	34	Logical Error	Use average = sum / 10.0;.
4	%d used to print a float variable	36	Compile-Time Warning	Use printf("Average = %.2f\n", average);.
5	Logic for finding highest marks is incorrect	48	Logical Error	Replace if(marks[i] < highest) with if(marks[i] > highest).
6	attendance[5] is out of bounds	64	Runtime Error	Valid indices are 0 to 4.
7	Loop condition i <= 5 accesses attendance[5]	68	Runtime Error	Replace with for(int i = 0; i < 5; i++).
8	Buffer overflow in strcpy(subject,"ComputerScience")	78	Runtime Error	Increase array size or use a shorter string.
9	String assigned directly to character array	86	Compile-Time Error	Use strcpy(branch, "IT");.
10	Character array college is not null-terminated	95	Runtime Error	Add college[5] = '\0';.
11	String "two" used as an array index	101	Compile-Time Error	Use an integer index, e.g., fees[2] = 50000;.
12	Floating-point value 3.5 used as an array index	103	Compile-Time Error	Use an integer index.
13	scanf() missing address operator &	121	Runtime Error	Use scanf("%d", &result[i]);.
14	Assignment operator = used instead of comparison operator ==	136	Logical Error	Replace with if(stock[2] == 300).

Sr No.	Error	Line Number	Error Type	Error Correction
15	Entire array assigned to another array	145	Compile-Time Error	Copy elements individually or use memcpy().
16	%d used to print a float value	153	Compile-Time Warning	Use printf("%.2f\n", salary[0]);.
17	Loop condition i <= 5 accesses inventory[5]	161	Runtime Error	Replace with for(int i = 0; i < 5; i++).
18	Buffer overflow in strcpy(city,"JaipurCity")	175	Runtime Error	Increase array size or use a shorter string.
19	Entire array assigned to another array	183	Compile-Time Error	Copy elements individually or use memcpy().
20	Missing semicolon after printf("Program Ended\n")	189	Compile-Time Error	Write printf("Program Ended\n");.

Question 3 :

Sr No.	Error	Line Number	Error Type	Error Correction
1	Integer division used for average calculation	20	Logical Error	Use average = total / 5.0;.
2	Logic for finding lowest marks is incorrect	42	Logical Error	Replace if(marks[i] > lowest) with if(marks[i] < lowest).
3	Loop condition i <= 5 accesses attendance[5]	61	Runtime Error	Replace with for(int i = 0; i < 5; i++).

Sr No.	Error	Line Number	Error Type	Error Correction
4	Buffer overflow in strcpy(subject,"Programming")	71	Runtime Error	Increase array size or use a shorter string.
5	Unnecessary & used with character array in scanf()	79	Warning	Use scanf("%s", branch);.
6	fees[5] is out of bounds	92	Runtime Error	Valid indices are 0 to 4.
7	Assignment operator = used instead of comparison operator ==	104	Logical Error	Replace with if(stock[2] == 30).
8	Entire array assigned to another array	113	Compile-Time Error	Copy elements individually or use memcpy().
9	%d used to print a float value	121	Compile-Time Warning	Use printf("%.2f\n", salary[0]);.
10	scanf() missing address operator &	129	Runtime Error	Use scanf("%d", &result[i]);.
11	String assigned directly to character array	145	Compile-Time Error	Use strcpy(city, "Jaipur");.
12	String "one" used as an array index	153	Compile-Time Error	Use an integer index, e.g., inventory[1] = 100;.
13	Floating-point value 2.5 used as an array index	155	Compile-Time Error	Use an integer index.
14	Loop condition i <= 5 accesses inventory[5]	165	Runtime Error	Replace with for(int i = 0; i < 5; i++).
15	Character array college is not null-terminated	180	Runtime Error	Add college[5] = '\0'; or declare char college[6];.

Sr No.	Error	Line Number	Error Type	Error Correction
16	Entire array assigned to another array	188	Compile-Time Error	Copy elements individually or use memcpy().
17	Buffer overflow in strcpy(course,"Computer")	196	Runtime Error	Increase array size or use a shorter string.
18	Loop condition i <= 5 accesses numbers[5]	204	Runtime Error	Replace with for(int i = 0; i < 5; i++).
19	Missing semicolon after return 0	218	Compile-Time Error	Write return 0;.
20	Program may exhibit undefined behavior due to multiple out-of-bounds array accesses	Multiple Lines	Runtime Error	Ensure all array indices remain within valid bounds (0 to 4).

Question 4 :

Sr No.	Error	Line Number	Error Type	Error Correction
1	Integer division used for average calculation	20	Logical Error	Use averageSales = totalSales / 5.0;.
2	%d used to print a float variable	22	Compile-Time Warning	Use printf("Average Sales = %.2f\n", averageSales);.
3	Logic for finding highest sale is incorrect	30	Logical Error	Replace if(sales[i] < highestSale) with if(sales[i] > highestSale).
4	monthlySales[5] is out of bounds	46	Runtime Error	Valid indices are 0 to 4.

Sr No.	Error	Line Number	Error Type	Error Correction
5	Loop condition <code>i <= 5</code> accesses <code>monthlySales[5]</code>	50	Runtime Error	Replace with <code>for(int i = 0; i < 5; i++)</code> .
6	Buffer overflow in <code>strcpy(product,"LaptopComputer")</code>	60	Runtime Error	Increase array size or use a shorter string.
7	String assigned directly to character array	68	Compile-Time Error	Use <code>strcpy(category, "Electronics");</code> .
8	String "first" used as an array index	76	Compile-Time Error	Use an integer index, e.g., <code>stock[0] = 100;</code> .
9	Floating-point value 2.7 used as an array index	78	Compile-Time Error	Use an integer index.
10	<code>scanf()</code> missing address operator <code>&</code>	96	Runtime Error	Use <code>scanf("%d", &orders[i]);</code> .
11	Assignment operator <code>=</code> used instead of comparison operator <code>==</code>	112	Logical Error	Replace with <code>if(revenue[2] == 3000)</code> .
12	Entire array assigned to another array	121	Compile-Time Error	Copy elements individually or use <code>memcpy()</code> .
13	<code>%d</code> used to print a float value	129	Compile-Time Warning	Use <code>printf("%.2f\n", commission[0]);</code> .
14	String assigned directly to character array	137	Compile-Time Error	Use <code>strcpy(city, "Jaipur");</code> .
15	Character array branch is not null-terminated	149	Runtime Error	Add <code>branch[4] = '\0';</code> .
16	Loop condition <code>i <= 5</code> accesses <code>employees[5]</code>	157	Runtime Error	Replace with <code>for(int i = 0; i < 5; i++)</code> .
17	Buffer overflow in <code>strcpy(department,"Marketing")</code>	171	Runtime Error	Increase array size or use a larger array.

Sr No.	Error	Line Number	Error Type	Error Correction
18	Entire array assigned to another array	179	Compile-Time Error	Copy elements individually or use memcpy().
19	Negative array index ratings[-1] is invalid	187	Runtime Error	Use only indices 0 to 4.
20	%s used to print an integer value feedback[0]	201	Runtime Error	Use printf("%d\n", feedback[0]);.
21	Missing semicolon after printf("Report Generated Successfully\n")	205	Compile-Time Error	Write printf("Report Generated Successfully\n");.

Question 5 :

Sr No.	Error	Line Number	Error Type	Error Correction
1	Integer division used for average calculation	20	Logical Error	Use average = total / 10.0;.
2	%d used to print a float variable	22	Compile-Time Warning	Use printf("Average = %.2f\n", average);.
3	Logic for finding highest marks is incorrect	30	Logical Error	Replace if(marks[i] < highest) with if(marks[i] > highest).
4	Logic for finding lowest marks is incorrect	42	Logical Error	Replace if(marks[i] > lowest) with if(marks[i] < lowest).
5	attendance[10] is out of bounds	58	Runtime Error	Valid indices are 0 to 9.
6	Loop condition i <= 10 accesses attendance[10]	62	Runtime Error	Replace with for(int i = 0; i < 10; i++).

Sr No.	Error	Line Number	Error Type	Error Correction
7	Buffer overflow in strcpy(subject,"DataStructures")	72	Runtime Error	Increase array size or use a shorter string.
8	String assigned directly to character array	80	Compile-Time Error	Use strcpy(department, "InformationTechnology");.
9	String "two" used as an array index	88	Compile-Time Error	Use an integer index, e.g., fees[2] = 50000;.
10	Floating-point value 4.5 used as an array index	90	Compile-Time Error	Use an integer index.
11	scanf() missing address operator &	108	Runtime Error	Use scanf("%d", &result[i]);.
12	Assignment operator = used instead of comparison operator ==	123	Logical Error	Replace with if(inventory[2] == 30).
13	Entire array assigned to another array	132	Compile-Time Error	Copy elements individually or use memcpy().
14	%d used to print a float value	140	Compile-Time Warning	Use printf("%.2f\n", salary[0]);.
15	String assigned directly to character array	148	Compile-Time Error	Use strcpy(city, "Jaipur");.
16	Character array branch is not null-terminated	160	Runtime Error	Add branch[4] = '\0';.
17	Loop condition i <= 5 accesses stock[5]	168	Runtime Error	Replace with for(int i = 0; i < 5; i++).

Sr No.	Error	Line Number	Error Type	Error Correction
18	Buffer overflow in strcpy(course,"Programming")	182	Runtime Error	Increase array size or declare a larger array.
19	Negative array index ratings[-1] is invalid	190	Runtime Error	Use indices from 0 to 4 only.
20	%s used to print an integer value feedback[0]	196	Runtime Error	Use printf("%d\n", feedback[0]);.
21	Entire array assigned to another array	204	Compile-Time Error	Copy elements individually or use memcpy().
22	temp[5] is out of bounds	216	Runtime Error	Valid indices are 0 to 4; use temp[4] or another valid index.